

Artificial Intelligence and Data Science

CSA1008 – Software Engineering

Assignment Title: CO3-AT2-Code Review & Repository Evaluation

Case Study Title: Smart Waste Collection System

(SDG 11: Sustainable Cities and Communities)

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1. Introduction

The current waste collection system often follows fixed schedules, where vehicles collect waste at predetermined times without considering the actual condition of bins. This can cause overflowing bins, unhygienic surroundings, unnecessary collection trips, and increased fuel costs. Fixed schedules are inefficient because some bins may be nearly empty while others may become full before the next scheduled collection.

A Smart Waste Collection System can solve this problem by monitoring bin conditions, identifying bins that require collection, and helping authorities plan efficient collection routes. This supports SDG 11 – Sustainable Cities and Communities by promoting cleaner, more efficient, and sustainable urban waste management.

2. Problem Analysis

Clearly identify:

- Overflowing bins
- Unnecessary collection trips
- High fuel consumption
- Poor route planning
- Difficulty monitoring collection activities

3. Proposed Solution

Explain the proposed system:

- Monitor bin status
- Identify bins requiring collection
- Generate collection requests
- Assign vehicles
- Optimize routes
- Track collection status
- Generate reports

4. Objectives

1. To monitor the status and fill level of waste bins efficiently.
2. To reduce overflowing waste bins and maintain cleaner surroundings.
3. To avoid unnecessary collection trips when bins are not full.
4. To optimize collection routes and reduce travel time.
5. To reduce fuel consumption and operational costs.
6. To track collection vehicles and activities effectively.

5. System Modules

Use a table:

Module	Function
Bin Monitoring	Monitors waste-bin status
Collection Management	Manages collection requests
Route Management	Plans efficient routes
Vehicle Management	Manages vehicles and assignments
Dashboard	Displays system information
Reporting	Generates collection reports

6. Functional Requirements

Include requirements such as:

- User login
- Monitor bins
- View bin status
- Create collection requests
- Assign vehicles
- Plan routes
- Update collection status
- View reports

7. Non-Functional Requirements

Requirement	Description
Performance	System should respond quickly
Reliability	System should operate consistently
Security	Only authorized users can access the system
Scalability	System should support more bins and vehicles
Usability	Interface should be easy to use
Maintainability	System should be easy to update

8. Actors and Use Cases

Actors:

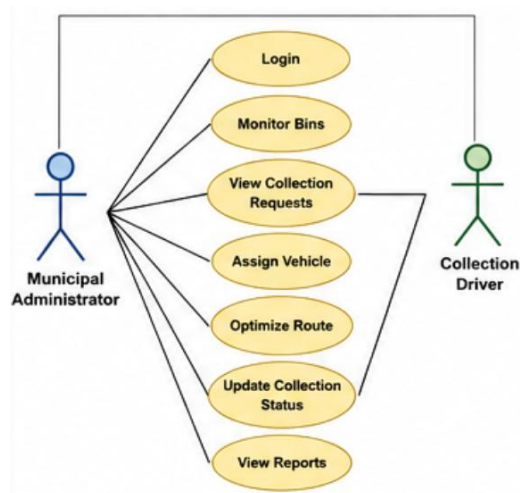
- Municipal Administrator

- Collection Driver
- Monitoring System

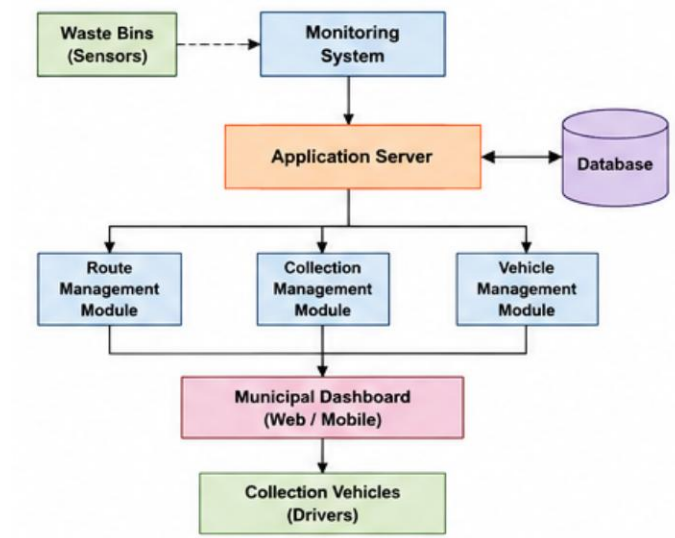
Use Cases:

- Login
- Monitor Bins
- View Collection Requests
- Assign Vehicle
- Optimize Route
- Update Collection Status
- View Reports

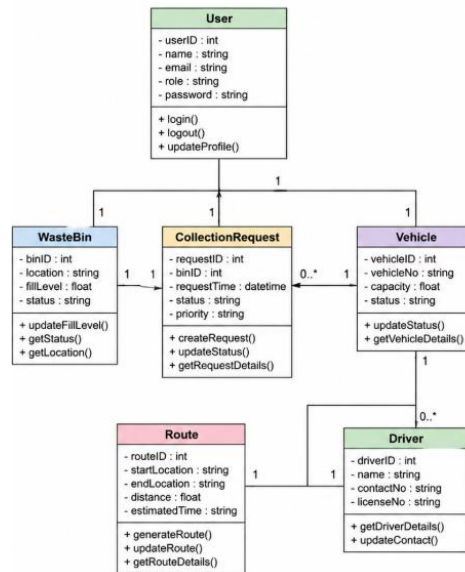
9. Use Case Diagram



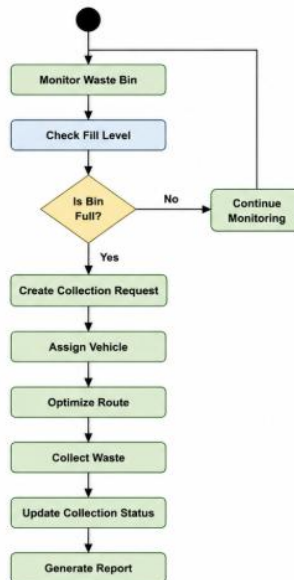
10. System Architecture Diagram



11. Class Diagram



12. Activity Diagram



13. Expected Benefits

- Reduced overflowing bins
- Reduced unnecessary trips
- Lower fuel consumption
- Better route planning
- Improved vehicle utilization
- Faster waste collection
- Better monitoring
- Improved urban cleanliness

14. Conclusion

The Smart Waste Collection System provides an efficient and sustainable solution for managing urban waste collection. By monitoring waste bins, optimizing collection routes, reducing unnecessary trips, and tracking collection activities, the system helps maintain cleaner surroundings while reducing fuel consumption and operational costs. Overall, the proposed system improves waste management efficiency, supports better use of municipal resources, and contributes to cleaner, safer, and more sustainable communities.